

Semantic-Adaptive Diffusion for Dynamic Spatiotemporal Fusion

Jinsong Zhang^{1,2} Ying Qu¹ * Yuan Liao^{1,2} Hairong Qi³ Zhenzhou Shao²

¹State Key Laboratory of Remote Sensing Science, Faculty of Geographical Science, Beijing Normal University. ²Beijing Key Laboratory of Light Industrial Robot and Safety Verification, Capital Normal University. ³Department of Electrical Engineering and Computer Science, University of Tennessee.

2231002005@cnu.edu.cn, yingqu@bnu.edu.cn, 2241002012@cnu.edu.cn, hqi@utk.edu, zshao@cnu.edu.cn

Abstract

Frequent and precise land surface monitoring is critical for numerous applications, yet existing satellites struggle to achieve both simultaneously. Spatiotemporal fusion (STF) tackles this challenge by integrating multiple satellite images to generate data with improved temporal and spatial resolution, enabling more frequent and precise land surface observations. However, current methods often fail to recover dynamic landscape changes due to significant scale discrepancies among multi-source images. To address these challenges, we propose a semantic-adaptive diffusion framework for dynamic spatiotemporal fusion (SA-STF), which constrains the solution space using low-resolution and high-frequency features decoupled via a Taylor-inspired decoder. By incorporating temporal feature alignment and semantic-adaptive fusion modules, SA-STF projects multimodal images with temporal dynamics into a unified latent space, and adaptively enhances spatial details while maintaining the spectral consistency of the reconstructed images. Experiments on benchmark datasets demonstrate that SA-STF outperforms existing methods in both quantitative and qualitative evaluations, particularly in complex and dynamic scenes.

1. Introduction

Satellites enable large-scale, repeated land surface observations, supporting various applications such as disaster emergency response [39], agricultural monitoring [6, 23], and smart city planning [31]. However, due to fundamental trade-offs imposed by sensor field-of-view and satellite orbital design [29], it is difficult to acquire imagery with both high temporal and high spatial resolutions. For example, satellites like MODIS, with 1–2 day revisit cycles and 250–1000 m spatial resolution, provide high temporal cov-

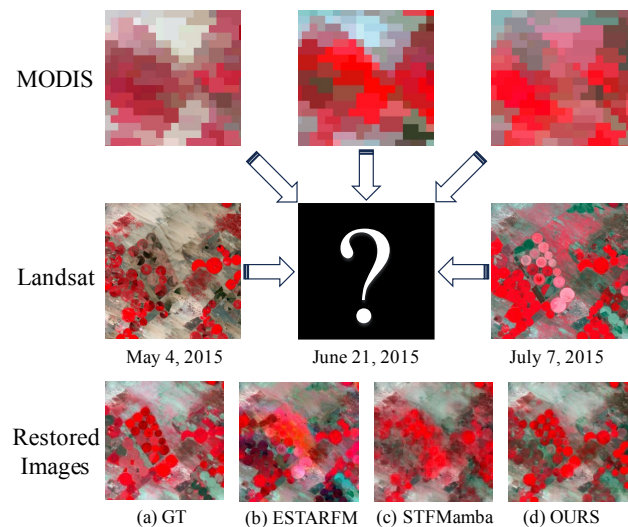


Figure 1. Spatiotemporal fusion. The first and second rows present the **coarse images** and **fine images** acquired by different satellites at different times. The goal is to reconstruct the unknown fine-resolution image. The third row shows the ground truth and restored images. (a) Ground Truth, (b) ESTARFM[41], (c) STFMamba[38], (d) Our method (SA-STF).

erage for regular crop monitoring but lack the spatial detail required for precise agricultural analysis. In contrast, Landsat offers a 30 m spatial resolution but has a low revisit frequency of approximately 16 days, limiting its ability to deliver timely observations. This spatiotemporal trade-off poses a major barrier for applications like precision agriculture, which require frequent and detailed monitoring [6, 23]. Spatiotemporal fusion (STF) addresses this challenge by integrating imagery from multiple satellites to generate continuously observed data with high spatiotemporal resolution, as illustrated in Fig. 1.

Existing spatiotemporal fusion (STF) methods can be broadly categorized into traditional and deep learning-based approaches. Traditional methods include those based on weighted functions [3, 41], unmixing [43], Bayesian infer-

*Corresponding author

ence [14], hybrid strategies [26, 42], and dictionary learning [8]. These approaches perform well when land-cover changes are limited, but struggle to capture complex dynamics over long temporal intervals, due to their reliance on linear models and hand-crafted features. To overcome these limitations, deep learning-based approaches have been introduced [34, 38], allowing automatic extraction of multi-scale features and model complex nonlinear relationships between coarse and fine images.

Despite their advantages, current deep learning methods still face three key challenges: 1) Undesirable artifacts. Most existing methods are purely data-driven and lack explicit constraints, making them prone to produce noticeable artifacts when fusing images with substantial scale discrepancies, especially in complex or dynamic regions. 2) Spectral distortion. Variations in acquisition time and phenological stages may lead to temporal inconsistencies in image sequences. Most existing methods do not explicitly model such temporal dynamics, resulting in spectral distortions in the reconstructed images. 3) Insensitive to semantic changes. Land-cover transitions, driven by natural dynamics or human activities, often lead to semantic changes across time. However, data-driven frameworks often fail to fully exploit the high-frequency information in reference images, restricting their ability to capture semantic changes induced by land-cover transitions.

To address the aforementioned challenges, we propose a semantic-adaptive diffusion framework for dynamic spatiotemporal fusion (SA-STF) to accurately recover land-cover changes. SA-STF bridges the multimodal scale gap through a residual diffusion mechanism guided by low-resolution observations and constraints. To mitigate undesirable artifacts, we introduce intermediate fusion features to explicitly model residual-noise interaction, based on which a decoder is designed to suppress noise via Taylor-based feature decomposition. To reduce spectral distortion caused by variations in acquisition time and phenological stages, a temporal feature alignment (TFA) module is proposed to project reference images into a unified latent space via linear regression. Furthermore, a semantic-adaptive fusion (SAF) module distinguishes changed and unchanged regions with semantic similarities and adaptively transfers high-frequency details from semantically matched patches, thereby reconstructing fine spatial structures in dynamic areas.

In summary, the main contributions of this paper are as follows:

- We propose a semantic-adaptive diffusion framework, which explicitly constrains the solution space using low-resolution observations and intermediate fusion features. A Taylor-inspired decoder is designed to improve reconstruction stability.
- A temporal feature alignment (TFA) module is proposed

to project reference images with temporal dynamics into a unified latent space to ensure the spectral consistency of the reconstructed images.

- A semantic-adaptive fusion (SAF) module is proposed to adaptively transfer high-frequency details from semantically matched regions to refine spatial details in dynamic areas.

2. Related Work

2.1. Spatiotemporal Fusion Methods

As discussed above, existing spatiotemporal fusion (STF) methods can be broadly categorized into six classes. Weighted function-based methods [3, 20, 37, 41], such as ESTARFM [41], predict target pixels through weighted reference observations but often perform poorly in mixed or heterogeneous regions. Unmixing-based methods [9, 11, 32, 43], such as LWU-STFM [11], reconstruct high-resolution details using linear mixing models. While they improve the representation of land-cover changes, they are prone to producing blocky artifacts. Bayesian-based methods [14, 33], such as STBDF [33], address the problem through probabilistic modeling, but rely heavily on high-quality input data. Dictionary learning-based methods [8, 21], such as SPSTFM [8], leverage sparse representations to learn the mappings transformation between low- and high-resolution images. Their performance is often limited due to their sensitivity to hyperparameters. Hybrid methods [4, 26, 42], such as FSDAF [42], integrate multiple of the aforementioned strategies but still struggle to capture complex land-cover dynamics.

Recently, deep learning-based methods have demonstrated advantages in spatiotemporal fusion by automatically extracting features and modeling complex nonlinear mappings between coarse and fine image pairs. For example, CNN-based STF [16, 19, 34] shows superior performance in restoring local details, while Transformer-based STF [1] effectively captures global low-frequency information. GAN-based approaches [18, 27, 36] enhance texture representation of fused images through generative mechanisms. Mamba-based STF [38] leverages the Mamba module to efficiently model the global spatiotemporal dependencies of remote sensing images. Despite numerous studies on STF, these methods are primarily data-driven, which often leads to undesired visual artifacts.

2.2. Diffusion Model

Diffusion models have shown remarkable success in natural image super-resolution tasks owing to their strong generative capabilities [25]. The denoising diffusion probabilistic model (DDPM) [7] is considered as a milestone for its improvement in image generation quality. Building on this, STFDiff [10] employs a conditional diffusion mechanism

to generate fused images, leveraging the detail reconstruction capability of DDPM. Subsequently, DDIM [28] improves generation efficiency by accelerating the sampling process through a non-Markovian formulation. In 2024, the residual denoising diffusion model (RDDM) [17] introduced both residual diffusion and noise diffusion processes to independently model image degradation and stochastic perturbations, thereby broadening the applicability of diffusion models.

3. Preliminaries

Given two coarse-fine image pairs at reference times (F_0, C_0) and (F_2, C_2) , along with a coarse image C_1 at the target time, the goal of STF is to reconstruct the corresponding fine image F_1 with conditional distribution $p(F_1 | Cond)$, where $Cond = \{C_1, F_0, C_0, F_2, C_2\}$. As the degraded image C_1 is already known, we introduce a residual diffusion process, in addition to the noise diffusion process, to prioritize certainty and guide the reverse image restoration [17]. During the training stage, the model leverages the predicted residuals x_{res}^θ and noise ε_θ to progressively reconstruct a high-resolution preliminary image F_1^t from its degraded version. During the inference stage, the image is rebuilt iteratively according to Eq. 2.

The framework consists of a forward degradation process, and a reverse reconstruction process, as illustrated in the supplementary materials.

$$\begin{aligned} x_t &= F_1 + \bar{\alpha}_t x_{res} + \bar{\beta}_t \varepsilon \\ x_{res} &= C_1 - F_1 \end{aligned} \quad (1)$$

where $\varepsilon \sim N(0, I)$, $t = 1, 2, \dots, T$, $\bar{\alpha}_t \in (0, 1)$ is the residual weighting factor, and $\bar{\beta}_t \in (0, 1)$ denotes the noise variance. Residuals and noise are progressively added to the target high-resolution image F_1 , and when t becomes sufficiently large, x_t can be approximated as a linear combination of noise, and the low-resolution observation C_1 .

In the reverse reconstruction process, unlike [17], where two distinct neural networks are used to independently estimate ε_θ and x_{res}^θ , our method introduces an intermediate variable F_1^t to reformulate the reverse diffusion process by establishing a linkage between ε_θ and x_{res}^θ using Eq. 1. In this way, only a single network is required for spatiotemporal fusion, thereby enhancing training stability and reducing the computational cost. The detailed derivation process is provided in the supplementary materials:

$$\begin{aligned} x_{t-1} &= \frac{\bar{\beta}_{t-1}}{\bar{\beta}_t} x_t + \gamma_t F_1^t + \lambda_t (C_1 - F_1^t) \\ \gamma_t &= 1 - \frac{\bar{\beta}_{t-1}}{\bar{\beta}_t} \\ \lambda_t &= \bar{\alpha}_{t-1} - \frac{\bar{\beta}_{t-1}}{\bar{\beta}_t} \bar{\alpha}_t \end{aligned} \quad (2)$$

where F_1^t refers to the preliminary output obtained by feeding x_t into the network at each time condition t , which is then used in Eq. 2 to iteratively generate the high-resolution target image. Since multimodal inputs differ in scale and are captured at different time points, SA-STF conducts multimodal fusion in the latent feature space, enabling better temporal alignment and maintaining spectral and semantic consistency in the reconstructed images. Although the noise and residual components are linearly combined as shown in Eq. 1, they become nonlinearly entangled in the feature space after propagating through the network. Inspired by [10], we use Taylor series expansion at the feature level to approximate the representation of x_t , enabling effective decoupling of noise and residual. Specifically, the feature extraction of x_t is defined as:

$$f_\theta(x_t) = f_\theta(F_1 + \bar{\alpha}_t x_{res} + \bar{\beta}_t \varepsilon) \quad (3)$$

By performing a first-order Taylor expansion around $x_t = F_1$, we obtain:

$$\begin{aligned} f_\theta(x_t) &= f_\theta(F_1) + \eta_{res} x_{res} + \eta_\varepsilon \varepsilon + o(F_1) \\ \eta_\varepsilon &= \bar{\beta}_t \nabla f_\theta(F_1) \\ \eta_{res} &= \bar{\alpha}_t \nabla f_\theta(F_1) \end{aligned} \quad (4)$$

where $f_\theta(\cdot)$ is a learnable feature extractor, and $o(F_1)$ denotes higher-order terms. $\nabla f_\theta(F_1)$ denotes a Jacobian matrix that serves as a local linear approximation of a nonlinear function. In the network design, $f_\theta(x_t)$ is extracted with a noise encoder, $\eta_{res} x_{res}$ is estimated with a fusion encoder $f_{\theta_e}(\cdot)$. According to Eq. 4, subtracting the residual $\eta_{res} x_{res}$ from $f_\theta(x_t)$ yields a combination of clean and noisy features. Then, a denoising decoder is introduced to remove noise and generate the intermediate variable F_1^t at each time point t . This deviation provides a solid foundation for a stable reverse reconstruction process. The detailed architecture is introduced in the following section.

4. Methodology

The overall architecture of the proposed SA-STF is illustrated in Fig. 2. The network comprises three main components: a noise encoder, a fusion encoder, and a Taylor-inspired decoder. The fusion encoder includes a shallow fusion module and a deep fusion module. The latter one integrates a temporal feature alignment module (TFA), and a semantic-adaptive fusion module (SAF) to handle temporal and semantic changes driven by vegetation phenology and land-cover transitions. The Taylor-inspired decoder includes a deep residual removal module and a denoising decoder. Details of the network parameters are provided in the supplementary materials.

4.1. Network Architecture

The network takes the noisy image x_t with embedded residuals, a set of prior images $Cond$, a residual control factor

provides rich structural details to support the subsequent denoising process.

4.4. Deep Fusion

The large resolution discrepancies between fine and coarse images pose a challenge for purely data-driven approaches, which often fail to capture dynamic information effectively, leading to reconstruction degradation. Thus, we develop a deep feature fusion module, composed of a temporal feature alignment module and a semantic-adaptive fusion module, to reconstruct temporal and semantic dynamics.

Temporal Feature Alignment. Due to differences in acquisition time and phenological variations, multi-temporal images often exhibit inconsistent spectral characteristics. To address this, we introduce a temporal feature alignment (TFA) module to align coarse- and fine-resolution image features by projecting them into a unified latent space via linear regression. Note that, since multi-temporal feature variations are aligned at the pixel level, a linear regression is sufficient to model the dynamics and it is more stable during optimization [24]. As illustrated in Fig. Model(b), two TFA modules capture temporal variation patterns from coarse-resolution features and use them to align fine-resolution features F_0^5 and F_2^5 with C_1^5 , respectively. The alignment is performed by adaptively estimating the temporal dynamic coefficient a and the discrepancy coefficient b .

Before coefficient estimation, the input features are refined through self-attention blocks to enhance their discriminative capability [30]. For clarity, the following description of the feature alignment process uses F_0^5 , C_0^5 , and C_1^5 as illustrative examples. The regression coefficients are estimated through convolutional blocks, as defined below.

$$\begin{aligned} a_l &= f_a(\text{Concat}(C_1^5, C_0^5)) \\ b &= f_b(F_0^5 - C_0^5), \end{aligned} \quad (5)$$

where f_a and f_b represent the convolutional blocks, and Concat denotes the concatenation operation. Due to the substantial scale disparity between coarse- and fine-resolution images, directly applying the coefficients derived from coarse-resolution features to fine-resolution features may lead to alignment bias. To address this, an adaptation module with a cross-attention mechanism is introduced to establish semantic correspondences across scales for adaptive alignment, as described below:

$$a_h = g(a_l, C_0^5, F_0^5), \quad (6)$$

where $g(\cdot)$ denotes the cross-attention module. This module refines the temporal regression coefficients by capturing the semantic mappings between coarse- and fine-resolution features. The refined coefficients are then used to model the temporal variations of the fine-resolution images, as follows:

$$HR_0^5(i, j) = a_h(i, j) \times F_0^5(i, j) + b. \quad (7)$$

Through the TFA module, we obtain HR_0^5 and HR_2^5 , which are then fed into the semantic-adaptive fusion module to facilitate the reconstruction of semantic changes.

Semantic-Adaptive Fusion. To improve reconstruction accuracy in change regions and reduce interference from invariant areas during fusion, we propose a semantic-adaptive fusion module that adaptively transfers the high-frequency details from semantically matched regions to refine spatial details in dynamic areas.

In time-series data, if a land-cover region remains stable over an extended period, it can be inferred that no significant changes have occurred at intermediate timestamps. Based on this assumption, we define such regions as invariant, while the remaining areas are considered as potential change regions.

We compute the Euclidean similarities between feature maps to obtain $\{S^{hr}, S_0^{lr}, S_2^{lr}\}$, where S^{hr} denotes the similarity between F_0^5 and F_2^5 , and S_0^{lr}/S_2^{lr} measures the similarity between F_0^5/F_2^5 and C_1^5 , respectively. Based on these similarities, we construct a region-wise mask M that adaptively distinguishes between unchanged and changed areas:

$$M(i, j) = \begin{cases} 1 & \text{if } S^{hr} < \min(S_0^{lr}, S_2^{lr}) \\ 0 & \text{if } S^{hr} > \min(S_0^{lr}, S_2^{lr}), \end{cases} \quad (8)$$

when $M = 1$, the area is considered unchanged and is directly represented by the deep feature F_0^5 from the prior image F_0 ; otherwise, it is considered as a changed region. For changed regions, we compute similarity masks S_0^r and S_2^r by comparing high-resolution features HR_0^5 and HR_2^5 from the temporal feature alignment module with the current feature C_1^5 . The feature with the higher similarity is selected as the final representation for the changed area.

$$F_d = \begin{cases} M \times F_0^5 + (1 - M) \times HR_0^5 & \text{if } S_0^r < S_2^r \\ M \times F_0^5 + (1 - M) \times HR_2^5 & \text{if } S_0^r > S_2^r \end{cases} \quad (9)$$

The deep semantic feature F_d at the target timestamp, obtained from the SAF module, provides precise semantic guidance for image reconstruction.

4.5. Taylor-inspired Decoder

In order to avoid the error accumulation in the reconstruction of F_1^t caused by the nonlinear entanglement of noise and residual components in the feature space of x_t , we propose a Taylor-inspired decoder consisting of a De-Res module and a denoising decoder, based on Taylor series expansion. The decoder aims to effectively decouple the noise and residual in the feature space, ensuring the high-quality reconstruction of F_1^t .

Deep Residual Removal Module. To maintain spectral and semantic consistency in the reconstruction image, we conduct multimodal fusion in the latent feature space. Based on

Eq. 4, a deep residual removal module is developed to remove residuals in $f_\theta(x_t)$. As shown in Fig. 2(a), the module consists of two TE-Blocks, each containing two residual convolutional blocks and a multi-layer perceptron (MLP). Specifically, $\nabla f_\theta(F_1) x_{res}$ is computed by subtracting F_d , extracted from the deep fusion module, from C_1^5 , and the resulting feature is fed into the first TE-Block. It explicitly embeds the residual strength $\bar{\alpha}_t$ into $\nabla f_\theta(F_1)$ via an MLP, enabling accurate prediction of the residual component $\eta_{res} x_{res}$ in $f_\theta(x_t)$. Then, $\eta_{res} x_{res}$ is subtracted from $f_\theta(x_t)$ to eliminate the mixed residual features. The second TE-Block further incorporates the noise intensity $\bar{\beta}_t$ into the residual-removed features, which enhances the denoising and reconstruction accuracy of the subsequent decoder.

Denoising Decoder. The denoising decoder is designed to remove noise components from $f_\theta(F_1) + \eta_\varepsilon \varepsilon + o(F_1)$. Similar to the noise encoder, it consists of five residual convolutional blocks. Each block fuses features from the shallow fusion module and integrates the noise intensity $\bar{\beta}_t$ via a multi-layer perceptron (MLP) for effective denoising. At each stage of the decoder, the spatial resolution of the features is progressively doubled, while the number of channels is gradually reduced. This process yields an accurate intermediate variable F_1^t , which provides reliable guidance for the reverse denoising process.

4.6. Optimization and Implementation Details

To further minimize the discrepancy between the predicted outputs F_1^t and the ground truth F_1 , we incorporate perceptual loss [12], defined as

$$L_{per} = \frac{1}{C_i H_i W_i} \|\varphi_i^{vgg}(F_1) - \varphi_i^{vgg}(F_1^t)\|_2^2 \quad (10)$$

where $\varphi_i^{vgg}(\cdot)$ denotes the feature maps extracted from the i -th layer of a pre-trained VGG-16 network; C_i , H_i , and W_i denote the number of channels, height, and width of the corresponding feature maps, respectively.

The reconstruction loss L_{rec} and temporal feature alignment loss L_{time} are defined in Eqs. 11 and Eqs. 12, respectively:

$$L_{rec} = \frac{1}{CHW} \|F_1 - F_1^t\|_1 \quad (11)$$

$$L_{time} = \frac{1}{N} \|f_{C_1} - \hat{f}_{C_1}\|_2^2 \quad (12)$$

where (C, H, W) denotes the dimensions of F_1 , N represents the number of pixels in f_{C_1} , and $\|\cdot\|_1$, and $\|\cdot\|_2$ represent the L1 and L2 norms, respectively. The term f_{C_1} refers to the deep features extracted from the coarse image C_1 at the target time, and $\hat{f}_{C_1}$ is the corresponding prediction generated by the temporal feature alignment module.

The overall loss function integrates L_{rec} , L_{per} , and L_{time} , as defined in

$$L_{overall} = \lambda_{rec} L_{rec} + \lambda_{per} L_{per} + \lambda_{time} L_{time} \quad (13)$$

where the parameters λ_{rec} , λ_{per} and λ_{time} balance the trade-offs among those loss components. To effectively leverage the auxiliary function of the temporal feature alignment module while minimizing interference with optimization during early training, we adopt a warm-up strategy. This allows the network to initially focus on the primary objective and gradually activate the temporal modeling capability. The loss is defined as follows:

$$\lambda_{time} = \lambda_{lr} \times 0.5 \times \left(1 - \cos\left(\pi \times \min\left(\frac{e}{E}, 1\right)\right)\right) \quad (14)$$

where E is the total number of training epochs, and e refers to the current training epoch.

The diffusion time step T is set to 100, and the DDIM sampler [28] is employed for inference with 50 sampling steps. The model is trained for 200 epochs with a batch size of 4, using the Adam optimizer to minimize the loss function with hyperparameters $\lambda_{rec} = 1$, $\lambda_{per} = 0.01$, and $\lambda_{lr} = 0.0001$. The initial learning rate is set to 0.0001 and is dynamically reduced by half every 40 epochs.

5. Experimental Results

5.1. Dataset Construction and Model Evaluation

To validate the performance of the proposed method, we conduct experiments on three benchmark datasets collected from heterogeneous regions: the Coleambally Irrigation Area (CIA) in southern New South Wales [2], dominated by irrigated rice, the Lower Gwydir Catchment (LGC) [2] in northern New South Wales affected by a flooding event, and the Ar Horqin Banner (AHB) [15] in the Inner Mongolia Autonomous region, representing mixed farming and pastoral systems. These datasets cover diverse land cover types and capture dynamic phenology and semantic changes. Further information on the datasets and their training-testing partitioning is provided in the supplementary materials.

We compare our method with five state-of-the-art methods belonging to different categories: ESTARFM [41], MLFF-GAN [27], SwinSTFM [1], STFDiff [10], and STF-Mamba [38]. Quantitative evaluation is conducted using five metrics, i.e., correlation coefficient (CC), structural similarity index measure (SSIM) [40], erreur relative globale adimensionnelle de synthèse (ERGAS) [13], root mean square error (RMSE), and spectral angle mapper (SAM) [35]. Higher CC and SSIM values indicate better structural fidelity, while lower ERGAS, RMSE, and SAM values reflect reduced spatial discrepancies and spectral distortion. To provide a comprehensive view of the performance variability across different test sets, we also use boxplots to visualize the distribution of RMSE, SAM, and SSIM values over changed regions, as shown in the supplementary materials.

5.2. Evaluation and Analysis

CIA Dataset: The fusion results on the CIA dataset are shown in Fig. 3, with quantitative evaluations reported in Table 1. Most areas are covered by farmland and vegetation, characterized by variations in crop phenology, while the surrounding drylands and woodlands remain relatively stable. Notably, on the CIA dataset, the traditional method ESTARFM [41] outperforms several deep learning-based approaches. This is mainly because the time span between the reference and predicted images is short, resulting in limited land cover changes and enabling linear models to produce effective fusion results. Deep learning-based methods, such as STFMamba[38] and SwinSTFM [1], tend to produce blurred results due to the resolution gap between coarse and fine images. In contrast, although generative models like STFDiff [10] and MLFF-GAN [27] can generate images with details, they fail to reconstruct semantic changes due to the lack of constraints. Nevertheless, the proposed SA-STF model achieves the best performance. This is attributed to its semantic-adaptive diffusion framework, which possesses strong generative capability and can adaptively transfer high-frequency details. As a result, SA-STF effectively reconstructs semantic changes while preserving spectral fidelity.

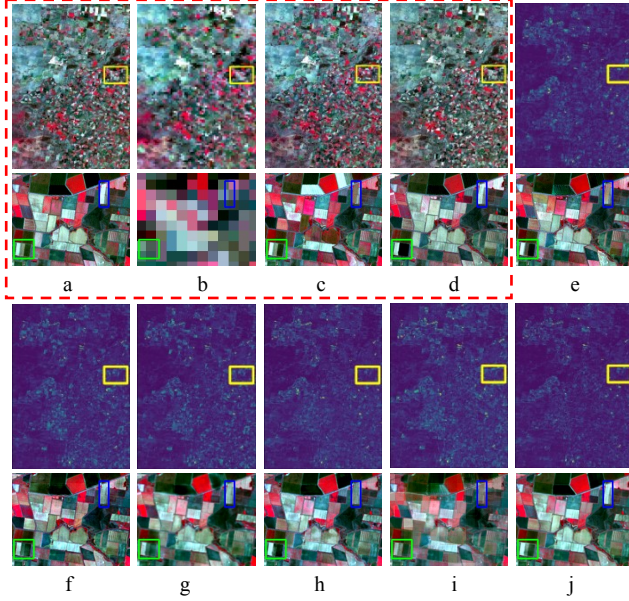


Figure 3. Reconstructed results on the CIA dataset. a. Ground truth (GT) Landsat image (LS) at t_1 b. MODIS image at t_1 c. Reference LS at t_0 d. Reference LS at t_2 e. ESTARFM[41] f. MLFF-GAN[27] g. SwinSTFM[1] h. STFDiff[10] i. STFMamba[38] j. OURS. The first row of (e)–(j) presents the SAM differences between the results and GT. Higher brightness indicates larger spectral errors.

LGC Dataset: Fig. 4 shows the fusion results on the

Table 1. Performance on the CIA dataset. The best results are highlighted in **bold** and the second best are underlined

Method	CC $\uparrow$	SSIM $\uparrow$	ERGAS $\downarrow$	RMSE $\downarrow$	SAM $\downarrow$
ESTARFM[41]	0.8823	<u>0.8885</u>	<u>0.9293</u>	<u>0.0222</u>	<u>0.0611</u>
MLFF-GAN[27]	0.8906	0.8669	0.9732	0.0236	0.0635
SwinSTFM[1]	<u>0.8910</u>	0.8821	1.0318	0.02228	0.0627
STFDiff[10]	0.8623	0.8729	1.1051	0.0242	0.0623
STFMamba[38]	0.8172	0.8605	1.1180	0.0265	0.0710
OURS	0.8950	0.8924	0.8910	0.0218	0.0538

LGC dataset, with the quantitative evaluation results reported in Table 2. In this flooding scenario, extensive vegetation loss leads to substantial land-cover changes between December 28, 2004, and January 13, 2005, whereas the changes between January 13, 2005, and January 29, 2005, are relatively minor. ESTARFM [41] performs poorly in such dynamic scenes. Compared to other deep-learning-based methods, STFMamba [38] performs better; however, its data-driven strategy restricts its ability to capture severe changes. In contrast, the proposed SA-STF leverages a temporal feature alignment module to align temporal dynamics, and a semantic-adaptive fusion module to restore land-cover transitions, thereby improving fusion accuracy.

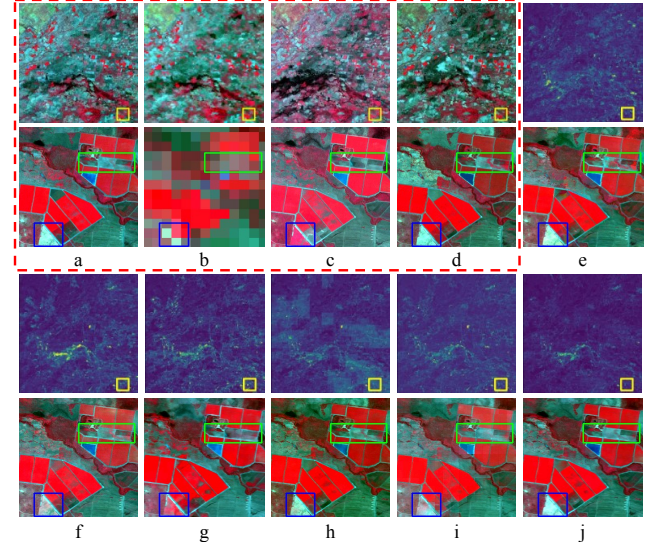


Figure 4. Reconstructed results on the LGC dataset. a. Ground truth (GT) Landsat image (LS) at t_1 b. MODIS image at t_1 c. Reference LS at t_0 d. Reference LS at t_2 e. ESTARFM[41] f. MLFF-GAN[27] g. SwinSTFM[1] h. STFDiff[10] i. STFMamba[38] j. OURS. The first row of (e)–(j) presents the SAM differences between the results and GT. Higher brightness indicates larger spectral errors.

AHB Dataset: Fig. 5 and Table 3 show the visual results and quantitative results on the AHB dataset, respectively. This region features diverse land cover types with distinct seasonal phenological dynamics, which in-

Table 2. Performance on the LGC dataset.

Method	CC↑	SSIM↑	ERGAS↓	RMSE↓	SAM↓
ESTARFM[41]	0.9150	0.9214	0.8564	0.0227	0.0651
MLFF-GAN[27]	0.9292	0.9376	0.7416	0.0191	0.0614
SwinSTFM[1]	0.9170	0.9278	0.7489	0.0193	0.0613
STFDiff[10]	0.9030	0.9281	0.9448	0.0192	0.0708
STFMamba[38]	0.9284	0.9420	0.6910	0.0174	0.0565
OURS	0.9360	0.9440	0.6751	0.0169	0.0531

creates the challenges of spatiotemporal fusion. Traditional methods like ESTARFM [41] fail to capture dynamic semantic changes. Although generative models like MLFF-GAN [27] and STFDiff [10] can reconstruct image details, they fail to preserve spectral information in change regions due to the absence of necessary constraints. In contrast, SA-STF effectively handles long-interval changes by aligning temporal dynamics and adaptively fusing features based on detected changes.

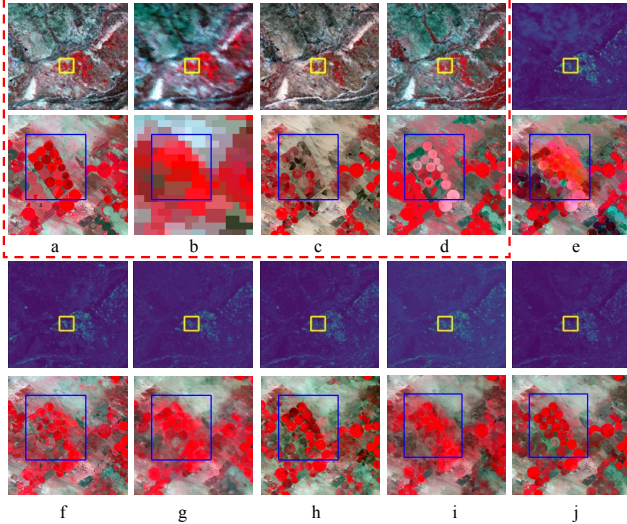


Figure 5. Reconstructed results on the AHB dataset. a. Ground truth (GT) Landsat image (LS) at t_1 . b. MODIS image at t_1 . c. Reference LS at t_0 . d. Reference LS at t_2 . e. ESTARFM[41]. f. MLFF-GAN[27]. g. SwinSTFM[1]. h. STFDiff[10]. i. STFMamba[38]. j. OURS. The first row of (e)–(j) presents the SAM differences between the results and GT. Higher brightness indicates larger spectral errors.

Table 3. Performance on the AHB dataset.

Method	CC↑	SSIM↑	ERGAS↓	RMSE↓	SAM↓
ESTARFM[41]	0.7654	0.8771	1.2403	0.0372	0.0951
MLFF-GAN[27]	0.8187	0.8826	0.9577	0.0316	0.0723
SwinSTFM [1]	0.8658	0.8830	1.0593	0.0326	0.0805
STFDiff[10]	0.8345	0.869	0.8808	0.0314	0.0764
STFMamba[38]	0.8612	0.8575	1.5199	0.0435	0.1065
OURS	0.8729	0.8962	0.8620	0.0283	0.0709

5.3. Ablation Study

To validate the contribution of each proposed component, we conduct ablation studies on the LGC dataset, as shown in Table 4. When only the Taylor-inspired decoder is retained, the results-except for the ERGAS—are comparable to RDDM [17], which is expected because residual estimation becomes less accurate without SAF and TFA, reducing the benefit of the Taylor structure. Incorporating either the temporal feature alignment (TFA) or the semantic-adaptive fusion (SAF) modules further boosts performance. When all components are integrated (OURS), the model achieves the best results on all five metrics, demonstrating its effectiveness in dynamic change recovery. Notably, even without the diffusion process (Only TFA+SAF), the model remains highly competitive, indicating that these modules can be adopted as plug-and-play components for spatiotemporal fusion. However, when the resolution gap between coarse and fine images is large, removing the diffusion mode causes a notable drop in ERGAS, and results in poorer recovery of fine spatial details. To evaluate SA-STF’s potential for downstream tasks, classification results are detailed in the supplementary material.

Table 4. Ablation study on the LGC dataset.

Method	CC↑	SSIM↑	ERGAS↓	RMSE↓	SAM↓
RDDM[17]	0.9194	0.9319	0.8504	0.0191	0.0683
Taylor-inspired framework	0.9194	0.9387	0.8486	0.0180	0.0652
+TFA	0.9262	0.9386	0.8083	0.0177	0.0621
+SAF	0.9271	0.9432	0.7230	0.0172	0.0562
OURS	0.9360	0.9440	0.6751	0.0169	0.0531
Only TFA+SAF	0.9306	0.9437	0.7045	0.0178	0.0545

6. Conclusion

In summary, we propose a semantic-adaptive diffusion for dynamic spatiotemporal fusion to recover land cover dynamics. It constrains the solution space with low-resolution observations and variables decoupled by a Taylor-inspired decoder, simplifying the reconstruction process while enhancing model stability and interpretability. By integrating a temporal feature alignment module and a semantic-adaptive fusion module, SA-STF effectively captures semantic changes while preserving the spectral consistency of the reconstructed images. Extensive experiments show that SA-STF consistently outperforms existing methods in both quantitative and qualitative evaluations. For rare abrupt changes without matched references, though SA-STF can preserve spectral trends of coarse observation using a learned data distribution, its capacity to reconstruct fine details may be limited. In future work we plan to incorporate additional prior knowledge to further improve the model’s robustness in these edge cases.

7. Acknowledgement

This work was supported by the National Natural Science Foundation of China under grant numbers 42293272, 42571096, and 42201368.

References

- [1] Guanyu Chen, Peng Jiao, Qing Hu, Linjie Xiao, and Zijian Ye. SwinSTFM: Remote Sensing Spatiotemporal Fusion Using Swin Transformer. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–18, 2022. [2](#), [6](#), [7](#), [8](#), [4](#)
- [2] Irina V. Emelyanova, Tim R. McVicar, Thomas G. Van Niel, Ling Tao Li, and Albert I.J.M. Van Dijk. Assessing the accuracy of blending Landsat–MODIS surface reflectances in two landscapes with contrasting spatial and temporal dynamics: A framework for algorithm selection. *Remote Sensing of Environment*, 133:193–209, 2013. [6](#)
- [3] Feng Gao, J. Masek, M. Schwaller, and F. Hall. On the blending of the Landsat and MODIS surface reflectance: predicting daily Landsat surface reflectance. *IEEE Transactions on Geoscience and Remote Sensing*, 44(8):2207–2218, 2006. [1](#), [2](#)
- [4] Zhuoning Gu, Jin Chen, Yang Chen, Yuean Qiu, Xiaolin Zhu, and Xuehong Chen. Agri-Fuse: A novel spatiotemporal fusion method designed for agricultural scenarios with diverse phenological changes. *Remote Sensing of Environment*, 299:113874, 2023. [2](#)
- [5] John A Hartigan and Manchek A Wong. Algorithm as 136: A k-means clustering algorithm. *Journal of the royal statistical society. series c (applied statistics)*, 28(1):100–108, 1979. [2](#)
- [6] Ze He, Shihua Li, Minghui Chang, Yuting Liu, Kaitong Liu, Lihong Wan, and Yong Wang. Novel harmonic-based scheme for mapping rice-crop intensity at a large scale using time-series sentinel-1 and era5-land datasets. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–23, 2024. [1](#)
- [7] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020. [2](#)
- [8] Bo Huang and Huihui Song. Spatiotemporal Reflectance Fusion via Sparse Representation. *IEEE Transactions on Geoscience and Remote Sensing*, 50(10):3707–3716, 2012. [2](#)
- [9] Bo Huang and Hankui Zhang. Spatio-temporal reflectance fusion via unmixing: accounting for both phenological and land-cover changes. *International Journal of Remote Sensing*, 35(16):6213–6233, 2014. [2](#)
- [10] He Huang, Wei He, Hongyan Zhang, Yu Xia, and Liangpei Zhang. STFDiff: Remote sensing image spatiotemporal fusion with diffusion models. *Information Fusion*, 111: 102505, 2024. [2](#), [3](#), [6](#), [7](#), [8](#), [4](#)
- [11] Xiaolu Jiang and Bo Huang. Unmixing-Based Spatiotemporal Image Fusion Accounting for Complex Land Cover Changes. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–10, 2022. [2](#)
- [12] Justin Johnson, Alexandre Alahi, and Li Fei-Fei. Perceptual losses for real-time style transfer and super-resolution. In *European conference on computer vision*, pages 694–711. Springer, 2016. [6](#)
- [13] Muhammad Murtaza Khan, Luciano Alparone, and Jocelyn Chanussot. Pansharpening quality assessment using the modulation transfer functions of instruments. *IEEE Transactions on Geoscience and Remote Sensing*, 47(11):3880–3891, 2009. [6](#)
- [14] Aihua Li, Yanchen Bo, Yuxin Zhu, Peng Guo, Jian Bi, and Yaqian He. Blending multi-resolution satellite sea surface temperature (SST) products using Bayesian maximum entropy method. *Remote Sensing of Environment*, 135:52–63, 2013. [2](#)
- [15] Jun Li, Yunfei Li, Lin He, Jin Chen, and Antonio Plaza. Spatio-temporal fusion for remote sensing data: an overview and new benchmark. *Science China Information Sciences*, 63(4):140301, 2020. [6](#)
- [16] Weisheng Li, Chao Yang, Yidong Peng, and Xiayan Zhang. A Multi-Cooperative Deep Convolutional Neural Network for Spatiotemporal Satellite Image Fusion. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14:10174–10188, 2021. [2](#)
- [17] Jiawei Liu, Qiang Wang, Huijie Fan, Yinong Wang, Yandong Tang, and Liangqiong Qu. Residual denoising diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2773–2783, 2024. [3](#), [8](#), [1](#)
- [18] Shuaijun Liu, Jia Liu, Xiaoyue Tan, Xuehong Chen, and Jin Chen. A Hybrid Spatiotemporal Fusion Method for High Spatial Resolution Imagery: Fusion of Gaofen-1 and Sentinel-2 over Agricultural Landscapes. *Journal of Remote Sensing*, 4:0159, 2024. [2](#)
- [19] Xun Liu, Chenwei Deng, Jocelyn Chanussot, Danfeng Hong, and Baojun Zhao. Stfnet: A two-stream convolutional neural network for spatiotemporal image fusion. *IEEE Transactions on Geoscience and Remote Sensing*, 57(9):6552–6564, 2019. [2](#)
- [20] Yuyang Ma, Yonglin Shen, Guoling Shen, Jie Wang, Wen Xiao, Huiyang He, Chuli Hu, and Kai Qin. STEPSBI: Quick spatiotemporal fusion with coarse- and fine-resolution scale transformation errors and pixel-based synthesis base image pair. *ISPRS Journal of Photogrammetry and Remote Sensing*, 206:1–15, 2023. [2](#)
- [21] Yidong Peng, Weisheng Li, Xiaobo Luo, Jiao Du, Xiayan Zhang, Yi Gan, and Xinbo Gao. Spatiotemporal reflectance fusion via tensor sparse representation. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–18, 2021. [2](#)
- [22] David MW Powers. Evaluation: from precision, recall and f-measure to roc, informedness, markedness and correlation. *arXiv preprint arXiv:2010.16061*, 2020. [2](#)
- [23] Peng Qin, Huabing Huang, Peimin Chen, Hailong Tang, Jie Wang, and Shuang Chen. Reconstructing ndvi time series in cloud-prone regions: A fusion-and-fit approach with deep learning residual constraint. *ISPRS Journal of Photogrammetry and Remote Sensing*, 218:170–186, 2024. [1](#)
- [24] Ying Qu, Zhenzhou Shao, and Hairong Qi. Non-local representation based mutual affine-transfer network for photoreal-

- istic stylization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7046–7061, 2021. [5](#)
- [25] Chitwan Saharia, Jonathan Ho, William Chan, Tim Salimans, David J Fleet, and Mohammad Norouzi. Image super-resolution via iterative refinement. *IEEE transactions on pattern analysis and machine intelligence*, 45(4):4713–4726, 2022. [2](#)
- [26] Wenzhong Shi, Dizhou Guo, and Hua Zhang. A reliable and adaptive spatiotemporal data fusion method for blending multi-spatiotemporal-resolution satellite images. *Remote Sensing of Environment*, 268:112770, 2022. [2](#)
- [27] Bingze Song, Peng Liu, Jun Li, Lizhe Wang, Luo Zhang, Guojin He, Lajiao Chen, and Jianbo Liu. MLFF-GAN: A Multilevel Feature Fusion With GAN for Spatiotemporal Remote Sensing Images. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–16, 2022. [2](#), [6](#), [7](#), [8](#), [4](#)
- [28] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020. [3](#), [6](#)
- [29] Enzhe Sun, Yongchuan Cui, Peng Liu, and Jining Yan. A decade of deep learning for remote sensing spatiotemporal fusion: Advances, challenges, and opportunities. *Information Fusion*, page 103675, 2025. [1](#)
- [30] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. [5](#)
- [31] Scott Workman and Armin Hadzic. Probabilistic image-driven traffic modeling via remote sensing. In *European Conference on Computer Vision*, pages 433–449. Springer, 2024. [1](#)
- [32] Mingquan Wu, Zheng Niu, Changyao Wang, Chaoyang Wu, and Li Wang. Use of modis and landsat time series data to generate high-resolution temporal synthetic landsat data using a spatial and temporal reflectance fusion model. *Journal of Applied Remote Sensing*, 6(1):063507–063507, 2012. [2](#)
- [33] Jie Xue, Yee Leung, and Tung Fung. A Bayesian Data Fusion Approach to Spatio-Temporal Fusion of Remotely Sensed Images. *Remote Sensing*, 9(12):1310, 2017. [2](#)
- [34] Mingzhu You, Xiangchao Meng, Qiang Liu, Feng Shao, and Randi Fu. CIG-STF: Change Information Guided Spatiotemporal Fusion for Remote Sensing Images. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–15, 2024. [2](#)
- [35] Roberta H Yuhas, Alexander FH Goetz, and Joe W Boardman. Discrimination among semi-arid landscape endmembers using the spectral angle mapper (sam) algorithm. In *JPL, Summaries of the Third Annual JPL Airborne Geoscience Workshop. Volume 1: AVIRIS Workshop*, 1992. [6](#)
- [36] Hongyan Zhang, Yiyao Song, Chang Han, and Liangpei Zhang. Remote sensing image spatiotemporal fusion using a generative adversarial network. *IEEE Transactions on Geoscience and Remote Sensing*, 59(5):4273–4286, 2020. [2](#)
- [37] Hua Zhang, Yue Sun, Wenzhong Shi, Dizhou Guo, and Nanshan Zheng. An object-based spatiotemporal fusion model for remote sensing images. *European Journal of Remote Sensing*, 54(1):86–101, 2021. [2](#)
- [38] Min Zhao, Xiaolu Jiang, and Bo Huang. Stfmamba: Spatiotemporal satellite image fusion network based on visual state space model. *ISPRS Journal of Photogrammetry and Remote Sensing*, 228:288–304, 2025. [1](#), [2](#), [6](#), [7](#), [8](#), [4](#)
- [39] Zhuo Zheng, Yanfei Zhong, Liangpei Zhang, Marshall Burke, David B Lobell, and Stefano Ermon. Towards transferable building damage assessment via unsupervised single-temporal change adaptation. *Remote Sensing of Environment*, 315:114416, 2024. [1](#)
- [40] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. [6](#)
- [41] Xiaolin Zhu, Jin Chen, Feng Gao, Xuehong Chen, and Jeffrey G. Masek. An enhanced spatial and temporal adaptive reflectance fusion model for complex heterogeneous regions. *Remote Sensing of Environment*, 114(11):2610–2623, 2010. [1](#), [2](#), [6](#), [7](#), [8](#), [4](#)
- [42] Xiaolin Zhu, Eileen H. Helmer, Feng Gao, Desheng Liu, Jin Chen, and Michael A. Lefsky. A flexible spatiotemporal method for fusing satellite images with different resolutions. *Remote Sensing of Environment*, 172:165–177, 2016. [2](#)
- [43] B. Zhukov, D. Oertel, F. Lanzl, and G. Reinhard. Unmixing-based multisensor multiresolution image fusion. *IEEE Transactions on Geoscience and Remote Sensing*, 37(3):1212–1226, 1999. [1](#), [2](#)